BRO Academy

Adaptive layout

Lesson

Lesson Plan

1
Introduction

2
Adaptive layout

3
Responsive layout

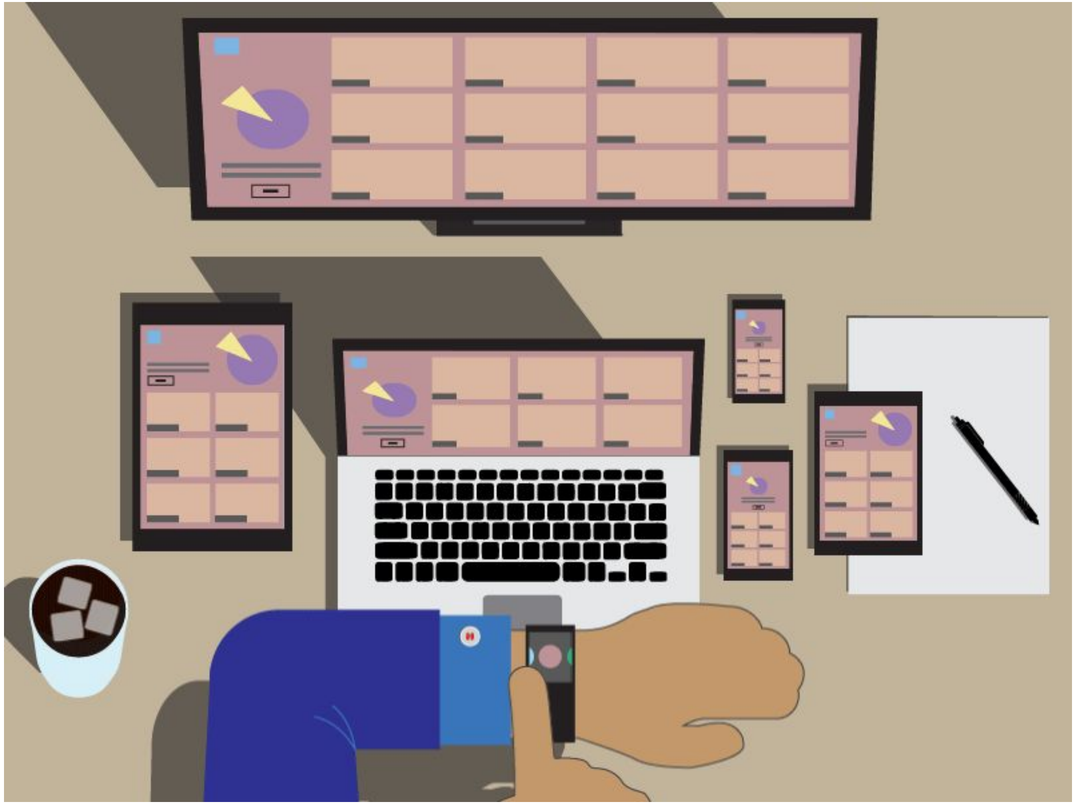
4
Tools and best practices

N

Devices

Adaptive markup

- **Adaptive markup** is a set of techniques that we use to make our site adapt to user’s device
- Additionally, it makes our web site look nice regardless of the content



Why is it important?

📌 Over 61.5% of global internet traffic comes from mobile devices

📌 Adaptive markup enhances UX

📌 Adaptiveness influences SEO

📌 Faster initial loading

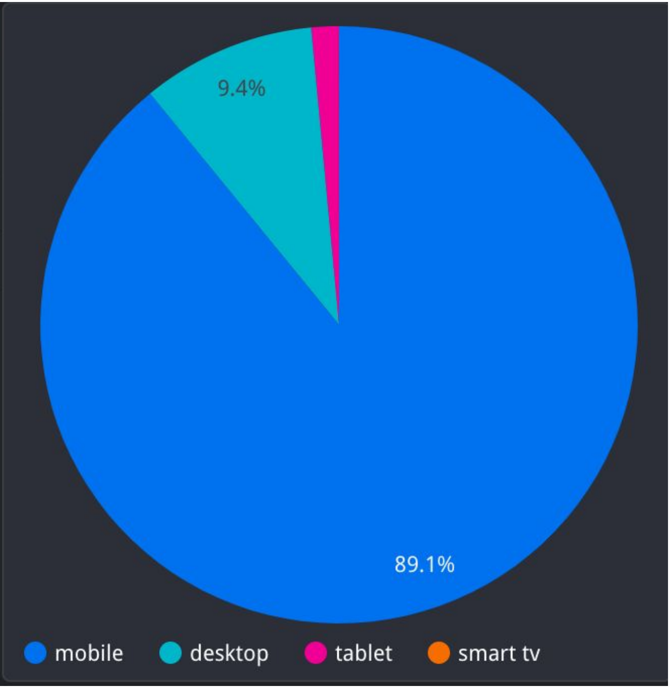
- LCP (Largest Contentful Paint)
- FCP (First Contentful Paint)
- FID (First Input Delay)
- CLS (Cumulative Layout Shift)

Mobile VS desktop

90% mobile
10% desktop

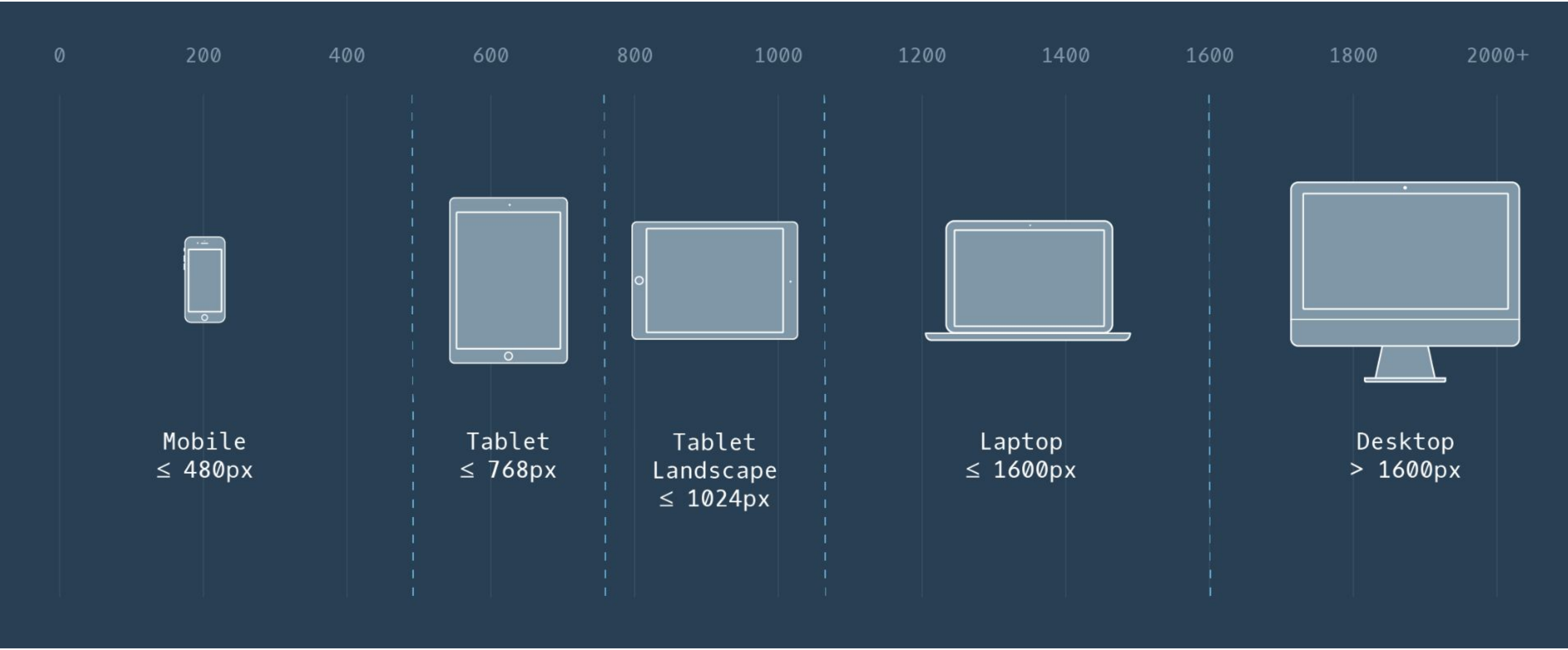
	Device...	Active...	Active...
1.	mobile	1,867,903	89.04%
2.	desktop	197,468	9.41%
3.	tablet	30,922	1.47%
4.	smart tv	329	0.02%

1 - 4 / 4



Android	VS	iOS
~56%	our company	~33%
~45%	worldwide	~18%

Screen sizes



Screen Resolution

< 375px

~20%

> 375px

~80%

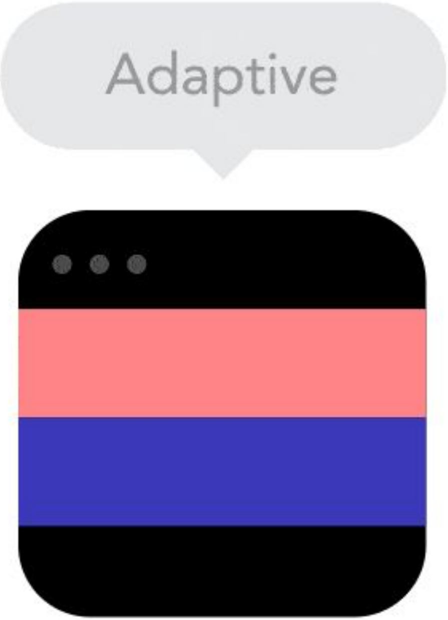
90% all devices are **360px-414px**

Responsive
vs
Adaptive

Responsive VS Adaptive

Responsive layout uses flexible grids and CSS to dynamically adjust content across all screen sizes.
https://adaptive-images.com/

Adaptive layout uses fixed layouts designed for specific screen sizes
https://lessframework.com/



Viewport meta tag

example →

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

💡 **width** - min viewport width

💡 **initial-scale** - zoom level on load

💡 **minimum-scale** - min zoom, default is 0.1

💡 **maximum-scale** - max zoom, default is 10

💡 **user-scalable** - defines if a site can be scalable; 0 - no, 1- yes (default)

https://codesandbox.io/p/sandbox/viewport-meta-tag-forked-yj5rk3

This page has a viewport width of 980 pixels. This is how it looks at 100% zoom on an HTC Desire.

The viewport is still 980px wide, but since the page is zoomed out it is squeezed into 480px, the resolution of the phone's screen.

Media queries

@media (condition) { /* CSS styles */ }

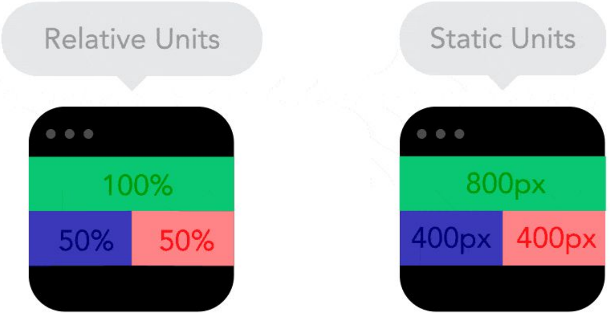
device's media type (print, screen, speech)	@media print @media screen	example
orientation (landscape, portrait)	@media (orientation: portrait)	example
aspect ratio	@media (max-aspect-ratio: 3/2)	
viewport size	@media (max-width: 768px) @media (min-width: 320px) and (max-width: 480px)	example
user preferences	@media (prefers-reduced-motion) @media (prefers-reduced-transparency)	example
screen resolution	@media (min-resolution: 192dpi)	
device pointer type	@media (pointer: coarse) @media (pointer: fine)	

example →

Responsive layout

✅ Using relative units

(%, em, rem, vw, vh)



✅ Flex container

✅ Grid container

Scss

```scss
$breakpoint-mobile: 360px;
$breakpoint-tablet: 677px;
$breakpoint-desktop: 1279px;
$breakpoint-desktop-large: 1727px;

@mixin mobile {
    @media (max-width: $breakpoint-mobile) {
        @content;
    }
}

@mixin tablet {
    @media (min-width: $breakpoint-tablet) {
        @content;
    }
}

@mixin desktop {
    @media (min-width: $breakpoint-desktop) {
        @content;
    }
}

@mixin desktop-large {
    @media (min-width: $breakpoint-desktop-large) {
        @content;
    }
}
```

How to use it

```scss
.contacts {
    display: flex;
    flex-direction: column;

    @include tablet {
        flex-direction: row;
    }
}
```

Adaptive images

Object-fit & Background-size

[object-fit MDN] & [background-size MDN]

- ✓ fill
- ✓ contain
- ✓ cover
- ✓ scale-down
- ✓ none

N

Picture

Picture

Adaptive images

[example →]

Srcset, Sizes & Picture tag

	`<img srcset sizes>`	`<picture>`
supports several image formats	❌	✅ via `<source>`
adapts to viewport width	✅ via `srcset` & `sizes`	✅ via `media` in `<source>`
supports retina/HDPI	✅ via `srcset`	✅ via `srcset` in `<source>`
optimization type	«initial load» optimized	layout-driven media selection

Bonus: responsive <iframe>

picture: source by type

```
<picture>
  <source srcset="corgi.avif" type="image/avif">
  <source srcset="corgi.webp" type="image/webp">
  <img src="corgi.jpg" width="100" height="100" alt="cute corgi">
</picture>
```

picture: source by screen width

```
<picture>
  <source srcset="mdn-logo-wide.png" media="(width >= 600px)" />
  <img src="mdn-logo-narrow.png" alt="logo" />
</picture>
<picture>
  <source media="(min-width:650px)" srcset="img_pink_flowers.jpg">
  <source media="(min-width:465px)" srcset="img_white_flower.jpg">
  <img src="img_orange_flowers.jpg" alt="Flowers">
</picture>
```

picture: source by screen orientation

```
<picture>
  <source
    srcset="images/examples/surfer.jpg"
    media="(orientation: portrait)" />
  <img src="images/examples/painted-hand.jpg" alt="" />
</picture>
```

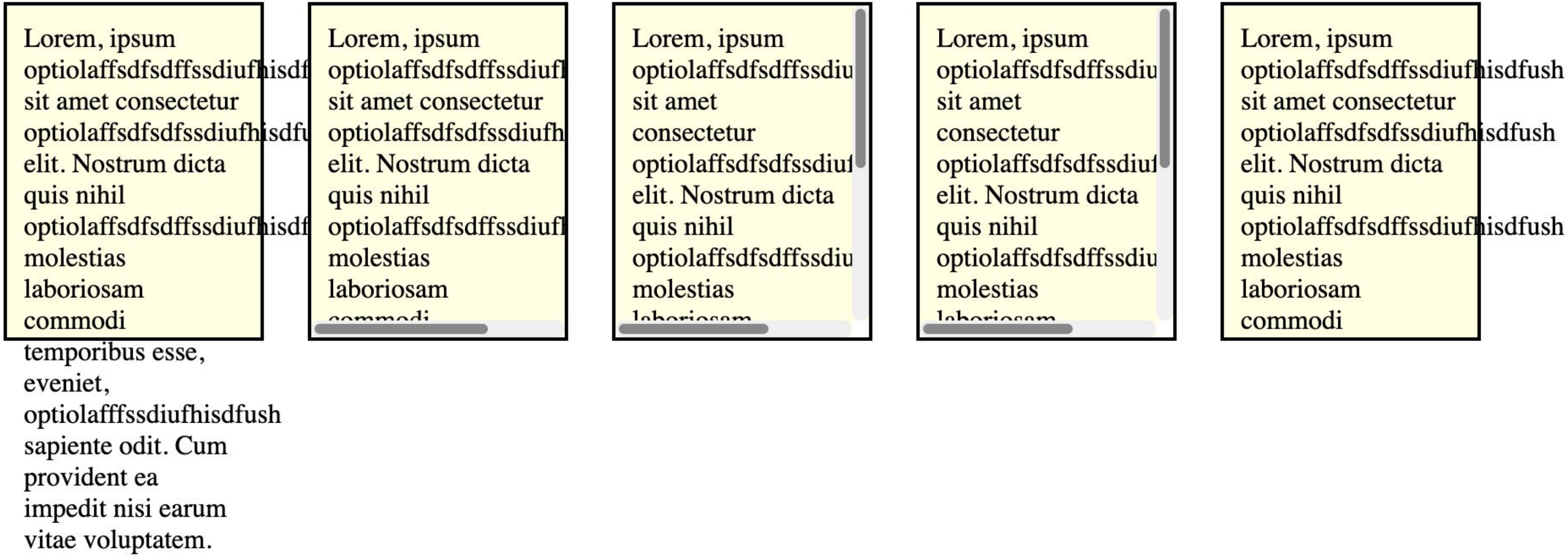
picture: source by screen resolution🔠

```
<picture>
  <source srcset="logo.png, logo-2x.png 2x" />
  <img src="logo.png" alt="logo" height="320" width="320" />
</picture>
```

Overflow

example →

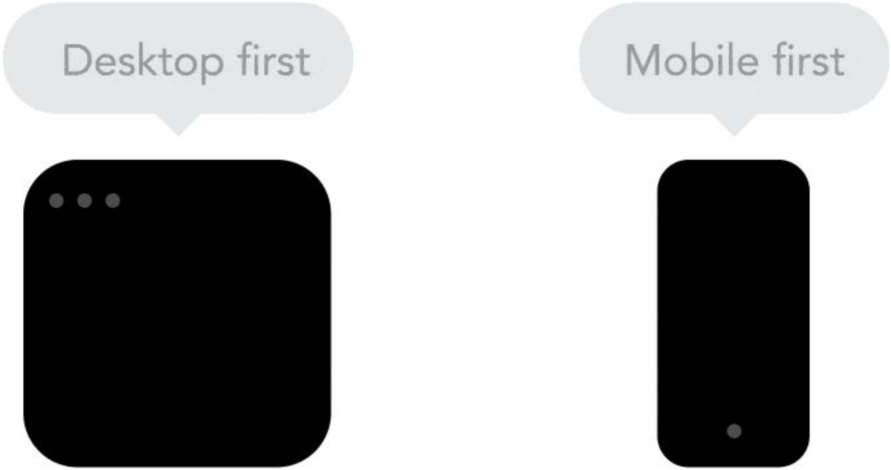
visible hidden scroll auto clip



Mobile-first vs. Desktop-first

Design starts with mobile screens, then scales up.
Focus on essentials → faster, cleaner, smarter UI.

- 📱 Prioritize core content
- 📱 Add features progressively
- 📱 Improve performance on mobile
- 📱 Boost SEO with mobile indexing
- 📱 Clean UX across all devices



Common mistakes

❌ **Website scales instead of adapting**	Always use <meta name="viewport" ...>
❌ **Disabled zoom**	Avoid user-scalable=no to allow user zoom if needed
❌ **Duplicated HTML for layouts**	Use one semantic structure, style via CSS/JS only
❌ **Too small elements on mobile**	Use at least 14–16px for body text and sufficient element size - 44×44px (Apple) 48dp (Google)
❌ **Hover-only interactions**	Menus and tooltips should work on tap/click instead of hover on mobile
❌ **Unoptimized images**	Don't load full-HD on mobile. Use **srcset**, **picture**, WebP
❌ **Testing only on desktop**	Test on real mobile devices, or use emulators (e.g., BrowserStack)
❌ **Improper use of vw/vh**	100vh may cut off content 100vw + scrollbar = horizontal scroll

How to test and fix overflows

example →

Tool	What it does
💡 **DevTools (Chrome, Firefox, Safari)**	Emulate screen sizes, pixel density, orientation; test breakpoints
💡 **Mobile OS Emulators**	iOS Simulator, Android Emulator for OS-specific bugs
💡 **BrowserStack / LambdaTest**	Cloud-based cross-device/browser testing, localhost support
💡 **Remote Debugging**	Inspect real devices via Chrome or Safari
💡 **Lighthouse & Validators**	Audit font sizes, tap targets, performance & accessibility
💡 **Screenshot Tools**	Services like Screenfly or Am I Responsive for quick visual checks
💡 **IDE Integrations**	VSCode + BrowserSync = live preview on multiple screens
💡 **Console Tools**	Log viewport width, DPR, breakpoints `console.log(window.innerWidth)`

How to reduce code with twig macros

```
{% macro picture(classes, source, alt, width, height, fit) %}
    <picture>
        <source srcset="/src/assets/images/{{ source }}?format=webp" type="image/webp">
        <source srcset="/src/assets/images/{{ source }}?format=avif" type="image/avif">
        <img
                  class="{{ classes }}"
                  src="/src/assets/images/{{ source }}"
                  alt="{{ alt }}"
                  loading="lazy"
                  decoding="async"
                  width="{{ width }}"
                  height="{{ height }}"
                  style="object-position:{{ fit }}">
    </picture>
{% endmacro %}
```

```twig
{% import "macros/picture.twig" as ui %}
```

How to use macros

twig

```twig
{% import "macros/picture.twig" as ui %}
```

twig

```twig
{{ ui.picture(
    'rounded-full w-16 h-16',  # classes
    'avatar.jpg',              # source
    'User photo',              # alt
    64,                        # width
    64,                        # height
    'center'                   # fit
) }}
```

twig

```twig
{{ ui.picture(
    'w-full h-auto',
    'hero-banner.png',
    'Main site banner',
    1440,
    600,
    'top'
) }}
```

Quality Criteria for HTML Course

❤️Mandatory for passing the course

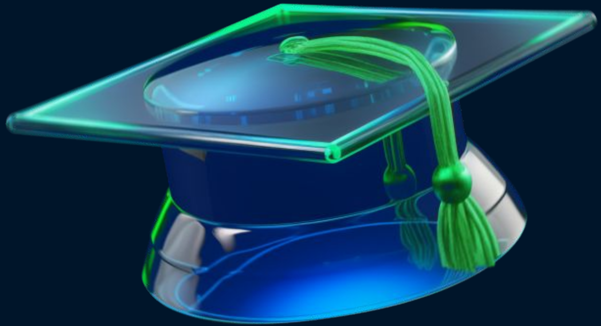
💛Required for the highest grade

💚Optional

- ❤️ Responsive adaptive layout is implemented.
- ❤️ No horizontal scroll on the full page.
- ❤️ Burger menu works on mobile.
- ❤️ Filter panel opens correctly on mobile.
- ❤️ Horizontal scroll is enabled for specific blocks.
- ❤️ Breakpoints are well chosen.
- ❤️ Consistent spacing and alignment.
- ❤️ Interactive elements are accessible on mobile.

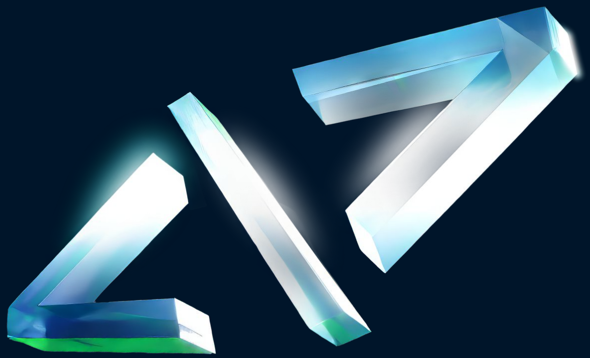
Summary

1. What is adaptive and responsive layout
2. What is media queries and their types
3. Adaptive images
4. Overflow
5. Mobile first approach
6. How to test and fix mistakes



Homework

1. Add adaptive layouts for all pages
2. Ensure the burger menu open and work correctly on mobile devices.



Test your layout on different screen sizes and pay close attention to spacing, alignment, and interactions.

Avoid horizontal scroll on the entire page — make sure the layout fits the viewport properly.

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QUESTIONS?

Please fill out the feedback form

It’s very important for us

THANK YOU!

Have a good evening!